Check for updates

Qizheng Liao
Department of Mechanical Engineering,
Beijing University of Posts
and Telecommunications,
Beijing 100088, P.R. China

J. Michael McCarthy
Department of Mechanical Engineering,
University of California, Irvine,
Irvine, CA 92697

On the Seven Position Synthesis of a 5-*SS* Platform Linkage

This paper builds on Innocenti's polynomial solution for the 5-SS platform that generates a one-degree of freedom movement through seven specified spatial positions of a rigid body. We show that his 60×60 resultant can be reduced to one that is 10×10. We then actuate the linkage using a prismatic joint on the sixth leg and determine the trajectory of the reference point through the specified positions. The singularity submanifold of this associated 6-SS platform provides information about the movement characteristics of the 5-SS linkage. [DOI: 10.1115/1.1330269]

1 Introduction

A rigid body supported by five legs, each with a ball joints at either end, moves with one degree of freedom. Let *S* represent the ball (or spherical) joint, then each leg is an *SS* chain, and the system is denoted as a 5-*SS* linkage. See Fig. 1. Chen and Roth [1] and Suh [2] introduced a design methodology for linkage systems of this type based on the numerical solution of the constraint equations for each supporting chain evaluated at the specified goal positions. Suh and Radcliffe [3] applied this methodology, known as *rigid body guidance*, to the design of a wide range of planar, spherical and spatial linkages. Innocenti [4] refers to this as the *spatial Burmester problem* acknowledging Burmester's approach to specifying the task of a planar 4*R* linkage using goal positions (Burmester [5]); see also Hall [6] or Denavit and Hartenberg [7]. This paper presents a new derivation for the polynomial that identifies the *SS* chains available for the 5-*SS* linkage, and a procedure for evaluating the performance of the resulting design.

1.1 The Design Process. The requirement that bar of the *SS* chain be the same length at each of seven goal positions yields seven quadratic equations in the coordinate vectors of the fixed and moving joints. It is known that there are at most 20 *SS* chains that solve this problem; see Chen and Roth [1] and Suh and Radcliffe [3]. Innocenti [4] presents an elimination procedure reduces the seven constraint equations to a 20th degree polynomial, which is the minimum degree polynomial possible.

Innocenti's derivation yields a 60×60 determinant in which the first 20 columns are linear in one unknown, and the remaining 40 columns are constants. He then uses an insightful numerical technique to obtain the coefficients of design polynomial as the solution to 21 linear equations. Our interest is in the continued symbolic analysis of this problem, and we note that Gaussian elimination of Innocenti's determinant can reduce it to a 20×20 determinant. In this paper, we present at two step elimination process that yields this polynomial as a 10×10 determinant.

1.2 Performance Evaluation. Given 20 available *SS* chains, we can assemble $C_{20}^5 = 15504$ 5-*SS* that reach the goal positions. In order to evaluate these designs we first consider the trajectory of the origin of the moving frame, which we call the *reference trajectory*. Often this trajectory consists of two or more independent circuits. If the goal positions are distributed on different circuits, then the 5-*SS* linkage is not useful because it cannot reach all the positions without being disassembled. We use a curve-tracing technique based on Newton's method in order to determine the reference trajectory.

Even if the linkage can reach all the goal positions, the linkage may ''jam'' because the actuator drives the system through a singular configuration. If all of the goal positions can be reached before the system hits a singularity, then linkage is said to be *input drivable*. The singularities of this system depend on the actuation scheme. For our purposes, we introduce a linear actuator that joins a moving pivot on one leg to the fixed pivot on another. In this way, we obtain a 5-5 in-parallel platform [8] in which one leg is actuated, (Fig. 3). Notice that there are 20 ways this 6th actuator can be added to a given 5-*SS* linkage. Each actuation scheme yields a different 5-5 platform.

The locus of singular configurations of 5-5 manipulator, termed the *singularity submanifold*, will intersect the reference trajectory in the singularities associated with the given actuation scheme. By evaluating the signs of the goal positions relative to the singularity submanifold, we can determine if a singularity must separate any two goal positions.

2 The Design Equations

Consider a reference frame *M* constrained to move relative to a fixed frame *F* by an *SS* chain. Denote the coordinate vector of the fixed joint as $\mathbf{B}=(B_x,B_y,B_z)^T$ and the moving joint as $\mathbf{P}=(P_x,P_y,P_z)^T$. As shown in Fig. 2. The position of the moving frame is defined by the 3×3 rotation matrix $[A]$ and the 3×1 translation vector $\mathbf{d}$, such that a point $\mathbf{x}$ in *M* has the coordinates $\mathbf{X}$ in *F* given by:

$$\mathbf{X}=[A]\mathbf{x}+\mathbf{d}. \tag{1}$$

It is easy to see that the length ρ of the *SS* chain is given by:

$$\rho^2=(\mathbf{P}-\mathbf{B})\cdot(\mathbf{P}-\mathbf{B})=([A]\mathbf{p}+\mathbf{d}-\mathbf{B})\cdot([A]\mathbf{p}+\mathbf{d}-\mathbf{B}). \tag{2}$$

This is the geometric constraint equation that characterizes the *SS* chain.

In order for the *SS* chain to be able to reach seven goal positions prescribed for the moving body *M*, the constraint (2) must be satisfied for each of the matrix-vector pairs $M_i:(A_i,d_i)$, $i=1,\ldots,7$. Thus, we have seven equations:

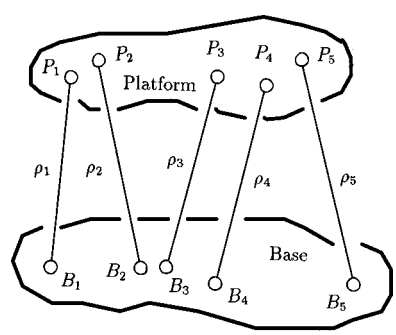

Fig. 1 A 5-*SS* spatial linkage

Contributed by the Design Automation Committee for publication in the JOURNAL OF MECHANICAL DESIGN. Manuscript received Nov. 1997. Associate Editor: B. Ravani.

 Copyright © 2001 by ASME

Downloaded from http://asmedigitalcollection.asme.org/mechanicaldesign/article-pdf/123/1/74/5797664/74_1.pdf?casa_token=KZGFIGj_dAgAAAAA:7uFgsbkAsrcdht4uds2i0cTZMXI23RwAekgtlm5GfirtfI3cMbVIG0MLwk5BeGdRR_TZ2oY by Ohio State University | OSU user on 24 February 2026

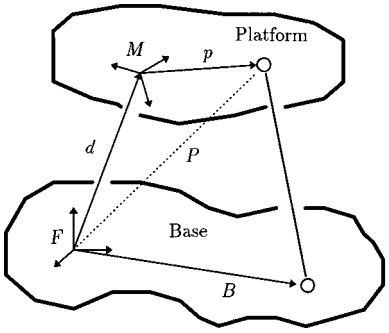

Fig. 2 An *SS* chain with fixed pivot $\mathbf{B}=(B_x,B_y,B_z)^T$ and moving pivot $\mathbf{P}=(P_x,P_y,P_z)^T$ in the fixed frame *F*

$$\begin{aligned}\rho^2&=([A_i]\mathbf{p}+\mathbf{d}_i-\mathbf{B})\cdot([A_i]\mathbf{p}+\mathbf{d}_i-\mathbf{B}),\\&=\mathbf{p}^2+\mathbf{d}_i^2+\mathbf{B}^2+2\mathbf{d}_i^T[A_i]\mathbf{p}-2\mathbf{B}^T[A_i]\mathbf{p}-2\mathbf{d}_i^T\mathbf{B},\\&\quad i=1,\ldots,7.\end{aligned}\tag{3}$$

Subtract the seventh of these equations from the first six to eliminate ρ^2, $\mathbf{p}^2$ and $\mathbf{B}^2$. The result is a set of six quadratic design equations for the *SS* chain:

$$\begin{aligned}\mathcal{Q}_i:\quad &2\mathbf{B}^T(-[A_i]+[A_7])\mathbf{p}+2(\mathbf{d}_i^T[A_i]-\mathbf{d}_7^T[A_7])\mathbf{p}\\&-2(\mathbf{d}_i^T-\mathbf{d}_7^T)\mathbf{B}+\mathbf{d}_i^2-\mathbf{d}_7^2=0,\quad i=1,\ldots,6.\end{aligned}\tag{4}$$

The unknowns in these equations are the six components of the two vectors **B** and **p** which define the locations of the fixed and moving pivots of the *SS* chain.

Notice that because the terms $\mathbf{p}^2$ and $\mathbf{B}^2$ have been canceled, the design Eq. (4) are linear in the components of **B** and **p**, when considered independently. Introduce the vectors $\mathbf{M}_i$ and scalars N_i that contain the components of **B**:

$$\begin{aligned}\mathbf{M}_i&=2(-[A_i]+[A_7])^T\mathbf{B}+2([A_i]^T\mathbf{d}_i-[A_7]^T\mathbf{d}_7),\\N_i&=2(\mathbf{d}_i^T-\mathbf{d}_7^T)\mathbf{B}+\mathbf{d}_i^2-\mathbf{d}_7^2,\quad i=1,\ldots,6,\end{aligned}\tag{5}$$

so (4) becomes:

$$\mathbf{M}_i^T\mathbf{p}+N_i=0,\quad i=1,\ldots,6.\tag{6}$$

We use these equations in a two step elimination procedure to obtain the 20th degree polynomial defining the desired *SS* chains.

2.1 Step 1: Solve for p. The first step in the elimination procedure is to consider the first four equations of the modified design Eq. (6):

$$\begin{bmatrix}\mathbf{M}_1^T & N_1\\ \mathbf{M}_2^T & N_2\\ \mathbf{M}_3^T & N_3\\ \mathbf{M}_4^T & N_4\end{bmatrix}\begin{Bmatrix}p_x\\p_y\\p_z\\1\end{Bmatrix}=[D_{1234}]\mathbf{p}=0.\tag{7}$$

This set of equations has a solution only if the determinant of the coefficient matrix $[D_{1234}]$ is zero, thus the components of **B** must satisfy the fourth degree polynomial:

$$\mathcal{D}_{1234}:\quad \det[D_{1234}]=0.\tag{8}$$

In fact, **B** must satisfy each of the 15 polynomials $\mathcal{D}_{ijkl}$ obtained in the same way for all combinations of four equations in (6) defined by $i<j<k<l=1,\ldots,6$.

For convenience we order the polynomials $\mathcal{D}_{ijkl}$ and relabel them $\mathcal{D}_j$, $j=1,\ldots,15$. Each of these polynomials has 35 terms and can be written as:

$$\mathcal{D}_j:\quad \sum_{i=1}^{35} a_{ji}B_x^lB_y^mB_z^n=0\quad l,m,n=0,\ldots,4,\tag{9}$$

$$l+m+n\leqslant 4,\quad j=1,\ldots,15.$$

We now seek to eliminate B_x and B_y in these equations to obtain a 20th degree polynomial in B_z.

2.2 Step 2: Eliminate B_x and B_y. At this point it is convenient to relabel the unknowns such that $x=B_x$, $y=B_y$ and $z=B_z$. Now rewrite (9) such that z is absorbed into the coefficients and only x and y appear explicitly, that is:

$$\begin{aligned}\mathcal{D}_j:\quad &b_{j1}x^4+b_{j2}x^3y+b_{j3}x^2y^2+b_{j4}xy^3+b_{j5}y^4+b_{j6}x^3+b_{j7}x^2y\\&+b_{j8}xy^2+b_{j9}y^3+b_{j10}x^2+b_{j11}xy+b_{j12}y^2+b_{j13}x+b_{j14}y\\&+b_{j15}=0\quad j=1,\ldots,15.\end{aligned}\tag{10}$$

Notice that because the total degree of any term is at most 4, we know: (i) the terms $b_{j1},\ldots,b_{j5}$ are constants; (ii) the terms $b_{j6},\ldots,b_{j9}$ are of degree 1 in z; (iii) the terms b_{j10}, b_{j11}, and b_{j12} are of degree 2; (iv) the terms b_{j13} and b_{j14} are of degree 3; and finally, (v) the term b_{j15} is of degree 4.

We now construct the eliminant of the 15 polynomials $\mathcal{D}_i$. Let **V** be the vector of 15 power products:

$$\mathbf{V}=(x^4,x^3y,x^2y^2,xy^3,y^4,x^3,x^2y,xy^2,y^3,x^2,xy,y^2,x,y,1)^T.\tag{11}$$

Assemble the coefficients of (10) into the 15×15 matrix,

$$[M_{15\times15}]=\begin{bmatrix}b_{1,1} & b_{1,2} & \cdots & b_{1,5} & b_{1,6} & \cdots & b_{1,15}\\ b_{2,1} & b_{2,2} & \cdots & b_{2,5} & b_{2,6} & \cdots & b_{2,15}\\ \vdots & \vdots & \ddots & \vdots & \vdots & & \vdots\\ b_{5,1} & b_{5,2} & \cdots & b_{5,5} & b_{5,6} & \cdots & b_{5,15}\\ b_{6,1} & b_{6,2} & \cdots & b_{6,5} & b_{6,6} & \cdots & b_{6,15}\\ \vdots & \vdots & \ddots & \vdots & \vdots & & \vdots\\ b_{15,1} & b_{15,2} & \cdots & b_{15,5} & b_{15,6} & \cdots & b_{15,15}\end{bmatrix},\tag{12}$$

so (10) becomes the matrix equation:

$$[M_{15\times15}]\mathbf{V}=0.\tag{13}$$

This equation has a solution only if the determinant of the coefficient matrix is zero, that is

$$\det[M_{15\times15}]=0.\tag{14}$$

Expansion of this determinant yields the desired 20th degree polynomial, because we have four first degree terms, three of degree two, two of degree three, and one of degree four which yield the sum 4+6+6+4=20.

The first five columns of constant terms in the determinant of $[M_{15\times15}]$ can be row reduced by Gaussian elimination to yield:

Downloaded from http://asmedigitalcollection.asme.org/mechanicaldesign/article-pdf/123/1/74/5797664/74_1.pdf?casa_token=KZGFIGj_dAgAAAAA:7uFgsbkAsrcdht4uds2l0cTZMXI23RwAekgtlm5GfirtfI3cMbVIG0MLwk5BeGdRR_TZ2oY by Ohio State University | OSU user on 24 February 2026

$$\det[M_{15\times 15}]=\begin{vmatrix} c_{1,1} & c_{1,2} & c_{1,3} & c_{1,4} & c_{1,5} & c_{1,6} & \cdots & c_{1,15} \\ 0 & c_{2,2} & c_{2,3} & c_{2,4} & c_{2,5} & c_{2,6} & \cdots & c_{2,15} \\ 0 & 0 & c_{3,3} & c_{3,4} & c_{3,5} & c_{3,6} & \cdots & c_{3,15} \\ 0 & 0 & 0 & c_{4,4} & c_{4,5} & c_{4,6} & \cdots & c_{4,15} \\ 0 & 0 & 0 & 0 & c_{5,5} & c_{5,6} & \cdots & c_{5,15} \\ 0 & 0 & 0 & 0 & 0 & c_{6,6} & \cdots & c_{6,15} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & 0 & 0 & c_{15,6} & \cdots & c_{15,15} \end{vmatrix}=0. \tag{15}$$

In general, the five constants, $c_{1,1}$, $c_{2,2}$, $c_{3,3}$, $c_{4,4}$, $c_{5,5}$, must be non-zero, for otherwise the determinant of $[M_{15\times 15}]$ is always zero. Thus, the polynomial defining the *SS* chains is given by the 10×10 determinant:

$$\det[M_{10\times 10}]=\begin{vmatrix} c_{6,6} & c_{6,7} & \cdots & c_{6,15} \\ c_{7,6} & c_{7,7} & \cdots & c_{7,15} \\ \vdots & \vdots & & \vdots \\ c_{15,6} & c_{15,7} & \cdots & c_{15,15} \end{vmatrix}=0. \tag{16}$$

This can be expanded using Mathematica on a Pentium PC computer.

2.3 Back Substitution. The expansion of the 10×10 determinant yields polynomial that is solved to determine the 20 values of z, for which (13) has a solution. For each of the roots $z_j = B_{z,j}$, $j=1,\ldots,20$, we solve (13) to obtain $\mathbf{V}_j$, which in turn yields $x_j=B_{x,j}$ and $y_j=B_{y,j}$. The result is 20 points $\mathbf{B}_j$ which are the fixed pivots of the *SS* chains.

For a given fixed pivot $\mathbf{B}_j$, use (6) to compute the associated moving pivot $\mathbf{p}_k$ measured in M. If M is in the ith position then we have $\mathbf{P}_{j,i}=[A_i]\mathbf{p}_j+\mathbf{d}_i$. The length of the *SS* bar joining these two pivots is ρ_j, which is obtained using (2).

3 The Reference Trajectory

The solution of the constraint equations above yields a set of *SS* chains that can reach the specified goal positions. Innocenti [4] presented seven goal positions for which all 20 *SS* chains can be determined. The result is that there are 15504 different 5-*SS* linkages that can reach these seven positions. An important concern is that the linkage pass through all seven positions in one continuous movement. In what follows, we describe the numerical curve-following strategy used to generate the trajectory of the origin of the moving frame, called the *reference trajectory*.

Consider the constraint Eq. (2) for each of the five legs of the linkage. Assemble them into the vector function, denoted $\mathbf{F}$, as follows:

$$\mathbf{F}=\begin{pmatrix} ([A]\mathbf{p}_1+\mathbf{d}-\mathbf{B}_1)\cdot([A]\mathbf{p}_1+\mathbf{d}-\mathbf{B}_1)-\rho_1^2 \\ \vdots \\ ([A]\mathbf{p}_5+\mathbf{d}-\mathbf{B}_5)\cdot([A]\mathbf{p}_5+\mathbf{d}-\mathbf{B}_5)-\rho_5^2 \end{pmatrix}. \tag{17}$$

The design of the *SS*-chains ensures that $\mathbf{F}([A],\mathbf{d})=0$ at the goal positions $M_i=([A_i],\mathbf{d}_i)$, $i=1,\ldots,7$. To generate the reference trajectory, we must determine the values of $([A],\mathbf{d})$ that satisfy this equation between the goal positions.

3.1 Curve Tracing. The reference trajectory is the continuous set of translation vectors $\mathbf{d}(t)$ locating the origin of the moving frame. Starting from any goal position, we step along this curve in increments of length d. Given the kth point $\mathbf{d}_k$ we search the points on the sphere of radius d about $\mathbf{d}_k$ for the one with a zero value of $\mathbf{F}$. Points $\mathbf{c}$ on this sphere are given by:

$$\mathcal{S}_k: \quad \mathbf{c}=\mathbf{d}_k+d\mathbf{s}, \tag{18}$$

where

$$\mathbf{s}=(\sin\alpha\cos\beta,\sin\beta,\cos\alpha\cos\beta)^T. \tag{19}$$

The angles α and β are longitude and latitude coordinates of the unit vector $\mathbf{s}$ measured relative to the fixed frame F. Newton's method is used to determine the values of these angles that locate $\mathbf{d}_{k+1}$.

To determine the zero of $\mathbf{F}$ at the $(k+1)^{\text{th}}$ increment, we must also compute the rotation matrix $[A_{k+1}]$. To specify this rotation, we use the three angles θ, ϕ, and ψ, defined such that:

$$[A]=[R_y(\theta)][R_x(-\phi)][R_z(\psi)], \tag{20}$$

where these matrices represent rotations about the y, x, and z axes respectively. The angles θ and ϕ can be visualized as the longitude and latitude of the z axis of M relative to F, and ψ defines the rotation of M about this axis.

3.2 Newton's Method. The movement of the platform is now defined by the parameter vector $\mathbf{r}=(\alpha,\beta,\theta,\phi,\psi)^T$, and we seek a root of the vector function $\mathbf{F}_{k+1}(\mathbf{r})$ defined on the sphere $\mathcal{S}_k$. Given the jth parameter value, $\mathbf{r}_j$, and the associated function value, $\mathbf{F}_{k+1}^j=\mathbf{F}_{k+1}(\mathbf{r}_j)$, we calculate $\mathbf{r}_{j+1}$, such that $\mathbf{F}_{k+1}^{j+1}$ approaches zero, as follows. Consider the linear approximation to $\mathbf{F}$ given by

$$\mathbf{L}(\mathbf{r})=\mathbf{F}_{k+1}^j+[D(\mathbf{r}_j)]\Delta\mathbf{r}, \tag{21}$$

where $[D(\mathbf{r}_j)]$ is the matrix of partial derivatives:

$$[D(\mathbf{r}_j)]=\left[\frac{\partial\mathbf{F}}{\partial\alpha},\frac{\partial\mathbf{F}}{\partial\beta},\frac{\partial\mathbf{F}}{\partial\theta},\frac{\partial\mathbf{F}}{\partial\phi},\frac{\partial\mathbf{F}}{\partial\psi}\right]. \tag{22}$$

This matrix was determined symbolically using Mathematica and will not be developed further here. A closely related matrix, the Jacobian of a six-legged parallel manipulator, is derived in detail in the next section.

The root of the linear function $\mathbf{L}(\mathbf{r})$ approximates that of $\mathbf{F}$, and provides the correction $\mathbf{r}_{j+1}=\mathbf{r}_j+\Delta\mathbf{r}$, given by:

$$\Delta\mathbf{r}=-[D(\mathbf{r}_j)]^{-1}\mathbf{F}_{k+1}^j. \tag{23}$$

Iteration of this equation yields the desired root $\mathbf{r}_{k+1}$, and therefore $\mathbf{d}_{k+1}$.

This routine can be started at any goal position and used to generate a circuit of the reference trajectory. Goal positions not on this circuit can be used as separate starting locations to generate other circuits.

4 The Singularity Submanifold

Our actuation scheme for the 5-*SS* linkage allows us to consider it as a 5-5 platform manipulator supported by six linear actuators, Fig. 3. Let $\overline{\rho}(t)=(\rho_1(t),\ldots,\rho_6(t))^T$ be the vector of actuator time histories chosen to drive this manipulator. Clearly, the movement of the 5-*SS* linkage is a subset of all movements available to this manipulator; and it occurs when $\dot{\rho}_i=0$, for i

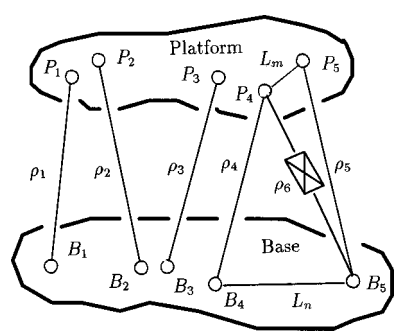

Fig. 3 A linear actuator connecting the base pivot B_n to moving pivot P_m can be used to actuate the 5-*SS* linkage

Downloaded from http://asmedigitalcollection.asme.org/mechanicaldesign/article-pdf/123/1/74/5797664/74_1.pdf?casa_token=KZGFIGj_dAgAAAAA:7uFgsbkAsrcdht4uds2i0cTZMXI23RwAekgtlm5GfirtfI3cMbVIG0MLwk5BeGdRR_TZ2oY by Ohio State University | OSU user on 24 February 2026

$=1,\ldots,5$. Any singular configurations that occur along this trajectory must be a subset of all the singular configurations of the 5-5 platform manipulator. In what follows, we derive this set of singular configurations, known as the *singularity submanifold* [9].

4.1 Derivative of the *SS* Constraint. Consider the constraint equation for the ith leg of the platform, which we assume, for the moment, can change length. The time-derivative of this constraint (2) yields:

$$2\rho_i\dot{\rho}_i = 2(\mathbf{P}_i - \mathbf{B}_i)\cdot\dot{\mathbf{P}}_i, \quad i=1,\ldots,6. \tag{24}$$

The velocities of the moving pivots $\dot{\mathbf{P}}_i$ can be determined from the movement of the platform by computing the time-derivative of the transformation Eq. (1) to obtain the well-known relation:

$$\dot{\mathbf{P}} = \vec{\omega}\times(\mathbf{P}-\mathbf{d}) + \dot{\mathbf{d}}, \tag{25}$$

where $\vec{\omega}$ is the angular velocity vector obtained from the skew-symmetric matrix $[\Omega]=[\dot{A}][A^{-1}]$. Substitute this into (24) to obtain

$$\rho_i\dot{\rho}_i = (\mathbf{P}_i - \mathbf{B}_i)\cdot(\vec{\omega}\times(\mathbf{P}-\mathbf{d}) + \dot{\mathbf{d}}), \quad i=1,\ldots,6. \tag{26}$$

This equation can be rewritten as the ''dot'' product of two six dimensional vectors:

$$\rho_i\dot{\rho}_i = \begin{pmatrix} \dot{\mathbf{d}} + \mathbf{d}\times\vec{\omega} \\ \vec{\omega} \end{pmatrix}^T \begin{pmatrix} \mathbf{P}_i - \mathbf{B}_i \\ \mathbf{P}_i\times(\mathbf{P}_i - \mathbf{B}_i), i=1,\ldots,6. \end{pmatrix}. \tag{27}$$

4.2 The Jacobian. The six-vector $\mathcal{P}_i = (\mathbf{P}_i - \mathbf{B}_i, \mathbf{P}_i\times(\mathbf{P}_i - \mathbf{B}_i))$ defines what is known as the *Plücker coordinate vector* of the line along the ith leg. We can go further and assemble these six equations into the single matrix equation:

$$Diag(\rho_i,\ldots,\rho_6)\dot{\bar{\rho}} = \begin{pmatrix} \dot{\mathbf{d}} + \mathbf{d}\times\vec{\omega} \\ \vec{\omega} \end{pmatrix}^T \times \begin{bmatrix} \mathbf{P}_1 - \mathbf{B}_1 \cdots \mathbf{P}_6 - \mathbf{B}_6 \\ \mathbf{P}_1\times(\mathbf{P}_1 - \mathbf{B}_1)\cdots\mathbf{P}_6\times(\mathbf{P}_6 - \mathbf{B}_6) \end{bmatrix}. \tag{28}$$

The matrix on the right side of this equation has as its columns the Plücker coordinate vectors of the six legs of the platform manipulator. It is the *Jacobian* $[J]$ of the system [10], that is:

$$[J] = \begin{bmatrix} \mathbf{P}_1 - \mathbf{B}_1 \cdots \mathbf{P}_6 - \mathbf{B}_6 \\ \mathbf{P}_1\times(\mathbf{P}_1 - \mathbf{B}_1)\cdots\mathbf{P}_6\times(\mathbf{P}_6 - \mathbf{B}_6) \end{bmatrix}. \tag{29}$$

When this matrix looses rank, the manipulator is said to be in a *singular configuration* [11,12].

For convenience in the following equations, introduce let the vector $\mathbf{u}_i = \mathbf{P}_i - \mathbf{B}_i$ denote the direction of the ith leg, and $\mathbf{v}_i = \mathbf{P}_i \times(\mathbf{P}_i - \mathbf{B}_i)$ be the moment of this vector about the origin of the fixed frame F. The condition that the manipulator is in a singular configuration is now given by:

$$\det[J] = \det\begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_3 & \mathbf{u}_4 & \mathbf{u}_5 & \mathbf{u}_6 \\ \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 & \mathbf{v}_4 & \mathbf{v}_5 & \mathbf{v}_6 \end{bmatrix} = 0. \tag{30}$$

It is not obvious, but this equation is defines a condition on the rotation $[A]$ and translation $\mathbf{d}$ of the platform. Notice that the locations of the fixed pivots $\mathbf{B}_i$ are given for a specific platform. Similarly, the locations $\mathbf{p}_i$ of the moving pivots in the platform are known constants. Thus, only the rotation matrix and translation vector in the transformation $\mathbf{P}_i = [A]\mathbf{p}_i + \mathbf{d}$ are variables in (30). The set of all rotations and translations that satisfy this equation is a subset of the configuration manifold of the manipulator system, called the *singularity submanifold*.

The equation of this submanifold is computed by expanding (30) into the sum of products of 3×3 determinants $|\mathbf{u}_{i_1},\mathbf{u}_{i_2},\mathbf{u}_{i_3}|$ and $|\mathbf{v}_{i_4},\mathbf{v}_{i_5},\mathbf{v}_{i_6}|$ using Laplace's formula. There are $C_6^3 = 20$ of these pairs, and the sign of each term is $(-1)^p$ where p is the number of transpositions in the permutation $(i_1,i_2,i_3,i_4,i_5,i_6)$ of the numbers $1,\ldots,6$. The equation of the singularity submanifold becomes [13]:

$$\det[J] = \sum_1^{20} (-1)^p |\mathbf{u}_{i_1},\mathbf{u}_{i_2},\mathbf{u}_{i_3}||\mathbf{v}_{i_4},\mathbf{v}_{i_5},\mathbf{v}_{i_6}| = 0. \tag{31}$$

4.3 Input-drivable. The condition that the 5-*SS* linkage be input-drivable is simply that it not pass through a singular position between two given goal positions. We check this by evaluating $\det[J(A_i,\mathbf{d}_i)]$ at each of the seven goal positions. The set of signs $(\text{sgn}(\det[J(A_1,\mathbf{d}_1)]),\ldots,\text{sgn}(\det[J(A_7,\mathbf{d}_7)])$ is called the *singularity signature* of the linkage for a given actuation scheme. If there is a sign change for two positions then the actuator must drive the system through a singularity as it moves along the reference trajectory between the positions. Thus, it is necessary that $\det[J(A_i,\mathbf{d}_i)]$ have the same sign for all M_i, $i=1,\ldots,7$. However, the singularity surface is complicated and may intersect the reference trajectory several times between goal positions. Thus, the lack of sign changes does not ensure that the system is input-drivable.

5 Numerical Example

In order to demonstrate our solution to the *SS* design equations and our approach to performance evaluation of 5-*SS* linkages, we use the seven goal positions specified by Innocenti [4], and listed in Table 1. These positions yield the 20 *SS* chains shown in Table 2, listed in order of increasing length of the connecting link. It took approximately 1.7 minutes to evaluate the Mathematica program on a Pentium 90 PC.

The shortest design. The first five *SS* chains combine to define the the shortest 5-*SS* linkage. For convenience, we label this as case [1,5,9,10,12]. The reference trajectory shown in Fig. 4, was obtained using the forward kinematics algorithm of the 5-5 platform obtain by connecting P_4 and B_5 with an actuator. This algorithm is described in Su et al. [8]. The range on the actuator length ρ_6 (Fig. 3) was determined such that:

$$\max(|\rho_5 - L_m|, |\rho_4 - L_n|) \leq \rho_6 \leq \min(\rho_5 + L_m, \rho_4 + L_n). \tag{32}$$

The figure required 500 evaluations at approximately 65 seconds each on a Pentium 90, for a run time of about 9 hours. This trajectory has three circuits. One of the goal position is on one loop, two of them on the second, and the remaining four are on the third circuit. This linkage can not reach all the goal position without disassembly, therefore it is not useful.

A continuous design. The curve tracing method described above was developed to provide a more efficient means to evaluate these linkages. Numerical experiments yielded the result that case [5,4,14] has a reference trajectory that passes continuously through all seven positions. Figure 5 is a schematic of this 5-*SS* linkage, and Fig. 6 shows its reference trajectory. The step size $d=0.05$ was used and required 1476 steps to return to the original position; at 3 seconds per step, the computation required about 1.25 hours.

Table 1 The seven goal positions for the 5-*SS* linkage taken from Innocenti [4]

Position	d_x	d_y	d_z	θ (°, y)	$-\phi$ (°, x)	ψ (°, z)
1	0	0	0	0	0	0
2	1	-0.7423	-0.1337	6.1802	4.27866	-97.9255
3	0.3182	-0.5085	-0.7922	-83.2605	-18.2345	73.6069
4	-0.1788	-1.7842	-1.0429	-170.031	39.5378	-50.9407
5	-1.258	0.8362	-1.4992	-84.7359	-29.1752	150.331
6	-3.5939	2.7283	-2.0334	-8.30385	5.0361	-68.2528
7	-0.0497	0.57	-1.4858	118.266	-33.7953	139.022

Downloaded from http://asmedigitalcollection.asme.org/mechanicaldesign/article-pdf/123/1/74/5797664/74_1.pdf?casa_token=kZGFIGj_dAgAAAAA:7uFgsbkAsrcdht4uds2l0cTZMXI23RwAekgtlm5GfirtfI3cMbVIG0MLwk5BeGdRR_TZ2oY by Ohio State University | OSU user on 24 February 2026

Table 2 The coordinates of the moving and fixed pivots of the 20 *SS* chains in order of increasing length

Solution	Length	p: moving pivot in M, B: fixed pivot in F.
1	$L = 2.861468869123672$	$\mathbf{p} = (1.360492865548295, 0.085024050292739, -1.028035710240964)$ $\mathbf{B} = (-1.453272348553658, -0.4131845167976985, -1.178094473901706)$
2	$L = 3.378687240499248$	$\mathbf{p} = (2.143531613579008, -0.9265607588914348, -0.1024823675124482)$ $\mathbf{B} = (-0.3764088193176859, -0.2693887825238974, -2.255051222913369)$
3	$L = 3.431210370415989$	$\mathbf{p} = (1.62608585637073, 1.834762099039148, -1.744217929714571)$ $\mathbf{B} = (-0.4050491297549506, -0.8842961363962122, -1.239819687749352)$
4	$L = 3.735060135162911$	$\mathbf{p} = (2.372954805082394, 0.6717656778172966, 0.2494500145792442)$ $\mathbf{B} = (0.8096238685269695, -0.9748667236421923, -2.716232424684218)$
5	$L = 4.28687477763693$	$\mathbf{p} = (-1.556531602549614, 1.152284417273162, 0.2325781719123078)$ $\mathbf{B} = (0.2606375321909033, 2.457661516918457, -3.424084558311483)$
6	$L = 4.393420153508743$	$\mathbf{p} = (-0.9780973930381817, 1.061517618102962, 0.4350433506205762)$ $\mathbf{B} = (0.9729226844203074, 2.906245345851284, -3.042404120310821)$
7	$L = 4.39656412131212$	$\mathbf{p} = (-0.6324234142575385, 4.071294209075877, 0.887601413696834)$ $\mathbf{B} = (1.2817499917712, 0.7174498241888071, -1.214170240508929)$
8	$L = 4.400578684327655$	$\mathbf{p} = (0.1609310274992683, -0.6353343948876942, 2.477586920801083)$ $\mathbf{B} = (-0.1483818121185966, 2.678908751814635, -0.4008172705422885)$
9	$L = 4.636146228591385$	$\mathbf{p} = (0.702611444609968, -0.3222595640489518, -0.6562393280429096)$ $\mathbf{B} = (-3.421031939770395, -0.2940958027682892, 1.462397944802235)$
10	$L = 7.944545162218186$	$\mathbf{p} = (0.8757286828225968, 2.477456027698051, 2.777190287466675)$ $\mathbf{B} = (-3.819906332138955, -3.725860842342321, 4.385191982323966)$
11	$L = 9.24978698702441$	$\mathbf{p} = (-0.3029258014375098, 0.2047331271874462, -0.9218467582432687)$ $\mathbf{B} = (-2.561160791457452, -4.157335184371265, -8.759658575355944)$
12	$L = 10.37494651388039$	$\mathbf{p} = (-1.343937644528796, -7.283121906676147, 5.128817218564246)$ $\mathbf{B} = (-4.070871351321328, -2.55934468371905, -3.696672936237892)$
13	$L = 10.98354769080216$	$\mathbf{p} = (3.458945827105294, 3.35244004302519, -9.663603320976998)$ $\mathbf{B} = (0.8993242062162834, -0.9070202037808347, 0.1314799068410434)$
14	$L = 13.40179957380535$	$\mathbf{p} = (0.3670416027161232, -0.5877795837745404, -0.9344681571987893)$ $\mathbf{B} = (-7.839270609239947, -0.08881660342599007, 9.649279177303846)$
15	$L = 15.8146455093244$	$\mathbf{p} = (-0.4713367468621585, 1.584099924841773, -1.981082504609068)$ $\mathbf{B} = (-7.735038584793455, -9.633021659617469, -10.43790410345379)$
16	$L = 25.7092626967865$	$\mathbf{p} = (-5.392505802672246, 3.202437873403068, 25.07939973887574)$ $\mathbf{B} = (0.07304601594400486, -0.5605758790383063, 0.2412504674558753)$
17	$L = 38.109139876478$	$\mathbf{p} = (5.480959038607537, -5.08897329709272, 14.71578734514883)$ $\mathbf{B} = (-3.244453485978457, -35.00665891818104, -7.218336878956598)$
18	$L = 75.96708416836856$	$\mathbf{p} = (-0.06790190416877504, 5.145360157455235, -4.516884900454201)$ $\mathbf{B} = (-48.95653552782445, -37.55354517763533, -43.98462948563325)$
19	$L = 86.06508698534856$	$\mathbf{p} = (51.35731251632376, 26.94308544748407, -62.18708229516659)$ $\mathbf{B} = (-7.966776371557833, 2.518284325360504, -4.817397220529439)$
20	$L = 193.3330871806713$	$\mathbf{p} = (-43.21571940083421, -113.3110520210707, -109.7202103700276)$ $\mathbf{B} = (75.5404901367107, 37.61219539925262, -87.430780516672)$

The next step was to consider the input-drivable condition, and we found that the singularity signature had sign changes for each of the twenty ways that a sixth actuator can be connected between the existing fixed and moving pivots. The thick segments of the trajectory in Fig. 6 have a positive Jacobian, and the thin segments are those for which it is negative, when the actuator connects the moving pivot of the first leg, P_1 to the base pivot of the second leg, B_2. It is easy to see that some of the goal frames lie on positive sections and others lie on negative sections, and the actuator must drive the linkage through a singularity as it moves.

The singularity signature for other designs. We considered computing the singularity signature for all of the available linkages. We started with the actuator joining the moving pivot P_1 to the base pivot B_2 and evaluated the 15504 possible linkages. The result was that only 124 had singularity signatures with no sign changes. The calculation required 2.5 seconds per linkage for a total runtime of about 11 hours. We did not check the 19 other actuation schemes.

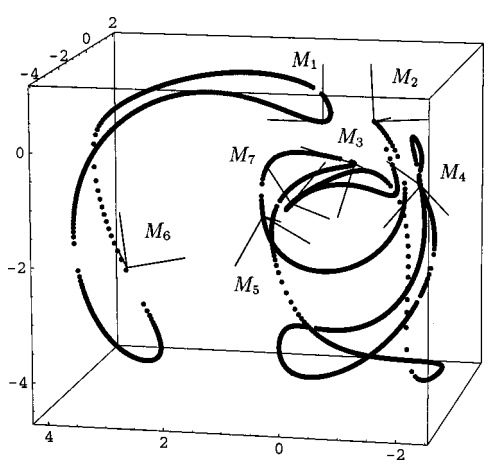

Fig. 4 The reference trajectory for the smallest 5-*SS* linkage with legs [1,5,9,10,12] consists of three circuits

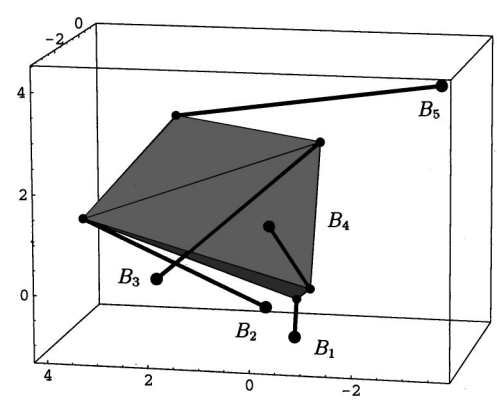

Fig. 5 The 5-*SS* chain with the *SS* chains [1,7,8,9,10] in Table 2 as legs

Downloaded from http://asmedigitalcollection.asme.org/mechanicaldesign/article-pdf/123/1/74/5797664/74_1.pdf?casa_token=KZGFIGj_dAgAAAAA:7uFgsbkAsrcdht4uds2i0cTZMXI23RwAekgtm5GfirtfI3cMbVIG0MLwk5BeGdRR_TZ2oY by Ohio State University | OSU user on 24 February 2026

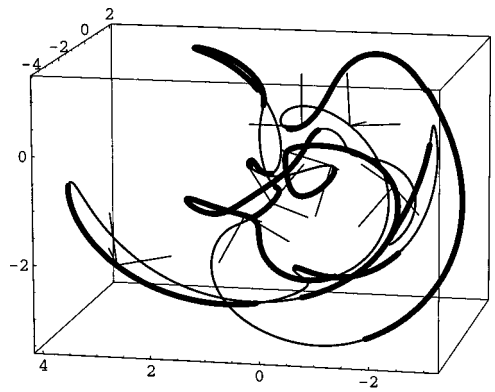

Fig. 6 The reference trajectory of the 5-*SS* linkage with legs [5,4,14,?,?] passes through all seven goal positions. The Jacobian is position on the thick section and negative on the thin sections

An interesting case that came out of this study is [5,10,12,14,15] which has no sign changes in its singularity signature, and the reference trajectory passes through all seven positions. See Fig. 7. Notice, however, that several of the goal positions lie on isolated segments in which the sign of the Jacobian is positive. This means there are actually two singularities between these goal positions. Thus, while we can be confident that a singularity signature with a sign change is undesirable, one without sign changes requires further study.

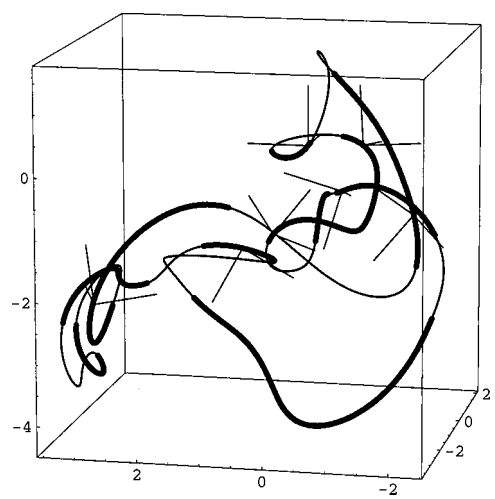

Fig. 7 The reference trajectory for the case [5,10,12,14,15] is continuous

6 Conclusion

This paper presents a new derivation of the polynomial that defines the *SS* chains that can reach seven specified goal positions, building on the work of Innocenti [4]. We then consider the evaluation of the resulting linkages. A curve tracing technique is used to provide an efficient means to check that a single assembly of the linkage can reach all the goal positions. We use a linear actuator to drive the linkage, and compute the sign of the Jacobian of the associated 5-5 in-parallel manipulator at each of the goal positions. Those linkages for which this singularity signature has a sign change must pass through a singularity between the goal positions. Finally, for those linkages that have no sign changes in their singularity signature, we examine the Jacobian at every point of the reference trajectory in order to ensure the system is input-drivable. The result is a procedure for the design and evaluation of 5-*SS* linkages that guide a rigid body through seven specified positions.

References

[1] Chen, P., and Roth, B., 1967, ''Design Equations for the Finitely and Infinitesimally Separated Position Synthesis of Binary Links and Combined Link Chains,'' ASME J. Eng. Ind., **91**, pp. 209–219.
[2] Suh, 1967.
[3] Suh and Radcliffe, 1978.
[4] Innocenti, C., 1994, ''Polynomial Solution of the Spatial Burmester Problem,'' ASME DE-Vol. 70, Mechanism Synthesis and Analysis.
[5] Burmester, L., 1886, *Lehrbuch der Kinematik*, Verlag Von Arthur Felix, Leipzig, Germany.
[6] Hall, 1961.
[7] Denavit and Hartenberg, 1964.
[8] Suh et al., 1997.
[9] Collins, C. L., and McCarthy, J. M., 1997, ''The Singularity Loci of Two Triangular Parallel Manipulators,'' *Proceedings of the 8th International Conference on Advanced Robotics*, Proc. R. Soc. London, Ser. A, Monterey, CA, July 6–8.
[10] Gosselin, C. M., and Angeles, J., 1990, ''Singularity Analysis of Closed Loop Kinematic Chains,'' IEEE Trans. Rob. Autom., **6**, No. 3, pp. 281–290.
[11] Merlet, 1989.
[12] Hall, Allen S., Jr., 1961, *Kinematics and Linkage Design*, Prentice-Hall, Englewood Cliffs, New Jersey.
[13] Mayer St-Onge and Gosselin, 1996.
[14] Kumar, V., 1992, ''Instantaneous Kinematics of Parallel-Chain Robotic Mechanisms,'' ASME J. Mech. Des., **114**, No. 3, pp. 349–358.
[15] Hartenberg, R. S., and Denavit, J., 1964, *Kinematic Synthesis of Linkages*, McGraw-Hill, New York.

Downloaded from http://asmedigitalcollection.asme.org/mechanicaldesign/article-pdf/123/1/74/5797664/74_1.pdf?casa_token=KZGFIGj_dAgAAAAA:7uFgsbkAsrcdht4uds2i0cTZMXI23RwAekgtm5GfirtfI3cMbVIG0MLwk5BeGdRR_TZ2oY by Ohio State University | OSU user on 24 February 2026